

Multi-View VLM Verification for Semantic Object Extraction from a Terrestrial Laser Scan Toward TOSM-Based Robot Environment Models

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Abstract—Semantic environment models such as the Triplet Ontological Semantic Model (TOSM) give indoor mobile robots the symbolic, explicit, and implicit knowledge needed for task-level autonomy, but populating them for an existing facility is still manual work. This paper presents an automated pipeline that extracts TOSM semantic objects from a single terrestrial laser scan and emits them as a VDA 5050-style JSON message. A deterministic stage segments the colored point cloud and fills the explicit model (pose, dimensions, color) with a provisional class from an open-vocabulary 3D classifier; a vision-language model (VLM) stage then verifies each class from multi-view object renders under forced tool-use and infers the implicit affordance model. We show that the obvious multi-view strategy — feeding all views into one prompt (early fusion) — *degrades* accuracy on sparse point renders, and propose per-view classification with confidence-weighted voting (late fusion) instead. On a 68.5M-point scan of a robot showroom (57 objects), late fusion cuts wrong labels from 24 (classifier-only) to 15 and is the strongest multi-view strategy (74% vs. 67% lenient accuracy over early fusion), while providing a free reliability signal: label votes unanimous across views are 94% correct, against 30% for split votes — gating validated when an on-site inventory check overturned a label that had fooled both the VLM and the human audit. Full-scene verification costs \$2.6.

Index Terms—multi-view object recognition, vision-language model, semantic mapping, terrestrial laser scanning, TOSM, VDA 5050

I. INTRODUCTION

Task-level autonomy for indoor mobile robots requires more than an occupancy grid: the robot must know *what* objects occupy the space, *where* they are, and *how* they can be interacted with. The Triplet Ontological Semantic Model (TOSM) [1] organizes this knowledge into a *symbolic* model (class, name, identity), an *explicit* model (pose, size, color), and an *implicit* model (affordances such as whether an object is a reliable landmark or can be displaced). TOSM-style databases drive navigation frameworks and multi-robot fleets, and their content maps naturally onto the JSON message conventions of VDA 5050 [2], the emerging interface standard between AGV fleets and master control systems.

In practice, the semantic-object database for a new facility is authored by hand. Terrestrial laser scanners (TLS) such as

the Leica BLK360 capture a dense, colored as-built model in minutes, and scan-to-BIM research [3] recovers structural geometry automatically, but object-level semantics remain the gap: open-vocabulary 3D classifiers such as Uni3D [4] are biased toward industrial structural classes — in our experiments office chairs were labeled *stair* or *machine* and displays *ladder* — and no point-cloud classifier produces the affordance-level knowledge TOSM needs.

Vision-language models (VLMs) offer a complementary signal: rendered views of a segmented object carry texture and context that a point-cloud embedding discards. Since multi-view recognition classically outperforms single-view, we render each object from four azimuths and study how a VLM should consume them. The naive strategy — all views in one prompt (*early fusion*) — turns out to *hurt*: sparse oblique point renders invite over-confident misreadings that contaminate the joint judgment. Classifying each view independently and aggregating by confidence-weighted vote (*late fusion*), the multi-view analogue of MVCCNN-style aggregation [5], recovers most of the lost reliability and exposes disagreement between views as an explicit uncertainty signal. The contributions are:

- an automated TLS-to-TOSM pipeline whose output is a VDA 5050-style `semanticObjects` message consumable by a fleet environment model;
- a schema-forced VLM verification protocol with per-view late fusion, plus prompt guardrails (size sanity, sparse-render caution, panel disambiguation) for point-render inputs;
- a six-configuration study on a real 68.5M-point BLK360 scan (57 objects) with a consistent, inventory-checked manual audit, showing early fusion degrades accuracy, late fusion is the strongest multi-view strategy, and vote unanimity yields 94%-precise gating.

II. RELATED WORK

Semantic environment models. TOSM [1] extends ontology-based robot knowledge bases [6] with the symbolic/explicit/implicit triplet; its records subsume the object fields of VDA 5050 [2] messages, which we adopt as the

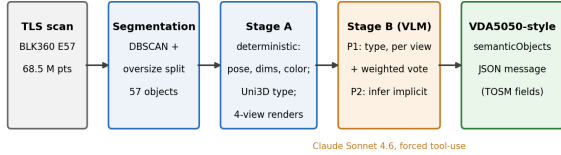


Fig. 1. Scan-to-semantics pipeline. Stage A deterministically fills the explicit TOSM model plus a provisional class; Stage B verifies the symbolic model per view with weighted late fusion and infers the implicit model.

output container; the taxonomy also underpins a knowledge-grounded 3D scene graph for LLM querying [7], whose records we populate automatically. **Scan-to-model.** As-built modeling from TLS has focused on structural elements [3]; object semantics are left to learned 3D segmentation [8] or open-vocabulary encoders such as Uni3D [4], which inherit training-corpus biases. **Multi-view recognition.** Aggregating rendered views is a classic route to 3D shape recognition (MVCNN [5]); we revisit the early-vs-late fusion question for VLM prompts over *sparse point* renders rather than mesh renders. **VLMs in robotics.** Recent systems use VLMs for open-set scene graphs [9] and language-driven maps [10]. We use a VLM in a narrower, auditable role: verifying one object at a time against measured geometry.

III. PIPELINE

A. Stage A: Deterministic Explicit Model

The input is a single registered BLK360 scan (E57, colored). After floor detection by RANSAC and wall removal, objects are segmented by DBSCAN [11] with a conservative re-clustering pass that splits clusters whose footprint exceeds 2.5 m (30→57 objects on our scene). Segmentation stays geometric because ScanNet-trained instance segmenters did not transfer to this industrial scene: on the same cloud, SPFormer assigned only 51% of points with furniture-biased labels, and PTv3 labeled 46% of points *wall* even after wall removal. For each object Stage A computes the explicit TOSM model directly from the points: bounding-box center (`poseX/Y`, `properties.poseZ`), PCA-based yaw (`poseTheta`), yaw-aligned extents (`dimensions`), and a mean RGB color name. Uni3D provides a *provisional* symbolic class, and four azimuth renders (0/90/180/270°, 512×512, point size 8, red 3D bounding box) are exported; smaller point sizes produced speckle-like views that misled both the VLM and the human auditors (Sec. IV-B). All Stage-A fields are measured, not generated; Stage B never modifies them.

B. Stage B: VLM Verification with Late Fusion

Stage B calls a commercial VLM (Claude Sonnet 4.6) with *forced tool-use*: the model must answer by calling a declared JSON tool, so every response is schema-valid by construction. Fig. 3 condenses the prompt design.

Prompt 1 (symbolic), once per view. The VLM sees ONE render, the measured dimensions, the provisional class, and

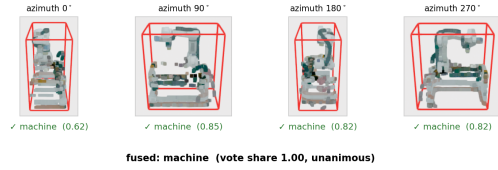


Fig. 2. Late fusion on one object (*machine_004*). Each of the four azimuth renders is classified independently; here all views agree, yielding the unanimous, confidence-1.0 case that our audit finds 94% reliable (Sec. IV-B).

P1 system (condensed). You are a semantic-mapping verifier for an indoor mobile robot. You see a render of ONE segmented 3D object (sparse laser points, red 3D box), its classifier label `uni3d_type`, and optional candidates. **Decision rule:** (1) if `uni3d_type` is correct, KEEP; (2) if wrong, replace — (a) prefer a fitting candidate, (b) else the most accurate common-noun label. Never ‘unknown’. **Size sanity:** measured dimensions are a HARD constraint; a cluster exceeding a class’s envelope is merged objects — label the dominant structure or ‘clutter’. **Sparse-render caution:** never claim parts not clearly outlined by points. **Panel rule:** thin flat objects are keyboard (key grid) / display (uniform dark face) / shelf — never ladder. **Chair gate:** requires visible seat pan 0.3–0.6 m high plus backrest. If uncertain, keep a size-compatible label. Respond ONLY via the `report_symbolic` tool.

P2 system (condensed). Infer implicit properties of ONE object from its views and confirmed type: `isKeyObject` (fixed, distinctive landmark), `isMovable`, `isOpen/canBeOpen` (doors only, else null). Respond ONLY via the `report_implicit` tool.

Fig. 3. Condensed Stage-B prompt design. Forced tool-use makes every response schema-valid JSON; measured geometry is read-only context.

an optional candidate vocabulary (*chair, TV, monitor, Mobile Robot, desk*). The decision rule is three-way: keep a correct label; if wrong, prefer a fitting candidate; else emit an open-vocabulary label. Guardrails address point-render failure modes (Fig. 3): dimensions as a hard constraint, oversize clusters treated as merged objects, thin-panel disambiguation, a ban on claiming parts not outlined by points, and a keep-or-clutter fallback when no label is clearly supported.

Late fusion. The four per-view reports are aggregated by confidence-weighted voting: each vote adds its confidence to its label’s score, the top-scoring label wins, and the final confidence is the winner’s share of the total. The per-view votes are stored in the record (`properties.viewVotes`), so a downstream consumer can distinguish a 4/4 unanimous label from a 2/4 split — an uncertainty signal that costs nothing extra. Fig. 2 shows a unanimous case: all four azimuths of a robot manipulator agree independently, giving fused confidence 1.0. Conversely, a single contaminated view is simply outvoted (a tall-backed chair whose frontal view read *Mobile Robot* fused to *chair* 3:1), whereas the same view inside one early-fusion prompt can drag the joint judgment toward the wrong label.

Prompt 2 (implicit), once per object. Given the fused type and all views, the VLM reports the TOSM implicit model: `isKeyObject`, `isMovable`, and (doors only) `isOpen/canBeOpen`.

TABLE I
TOSM $\rightarrow$ VDA 5050-STYLE JSON FIELD MAPPING

TOSM model	element	JSON field
symbolic	class	type
	name / identity	name, id
explicit	position (x, y)	poseX, poseY
	height z	properties.poseZ
	orientation	poseTheta
	size	dimensions{l, w, h}
	color	color
implicit	landmark / movability	properties.isKeyObject, isMovable, isOpen, canBeOpen
–	door state	
	VLM audit trail	properties.viewVotes, symbolicReason, confidence

C. TOSM to VDA 5050-Style Message

The completed records are serialized as one `semanticObjects` JSON message with a VDA 5050-style header. Table I shows the field mapping; TOSM fields with no standard slot travel in the extensible `properties` map, so the message remains readable by consumers that only know the base schema.

IV. EXPERIMENTS

A. Setup and Audit Protocol

We evaluate on a single BLK360 scan of an indoor robot showroom (68.5M colored points, 57 segmented objects) with Claude Sonnet 4.6. Six configurations are compared (Table II): the classifier alone; single-view and four-view early-fusion prompts, each with the initial sparse renders/base prompt and with dense renders/guardrail prompt; and four-view late fusion. To grade them consistently, two of the authors audited all 57 objects against the *dense* four-view renders and the measured dimensions, rating every configuration’s final label *correct*, *plausible* (consistent with the views but not verifiable from renders alone), or *wrong*. Metered cost: late fusion \$2.63 per scene (285 calls, \$0.046/object); early fusion \$1.27; single view \$0.87.

B. Multi-View Study

Table II and Fig. 5(b) summarize the audit. Three findings stand out.

(1) *Early fusion hurts*. Feeding four sparse views into one prompt dropped strict accuracy from 60% (single view) to 49%: extra oblique speckle views encouraged over-committed readings, e.g. two electrical cabinets confidently relabeled *chair*. Denser renders and the prompt guardrails did not close the gap (46% vs. 51%).

(2) *Late fusion is the strongest multi-view strategy*. Judging each view independently and voting beats early fusion on every measure (15 vs. 19 wrong labels, 74% vs. 67% lenient): a contaminated view can no longer dominate — it is outvoted. It does not, however, overtake the best single canonical view (60%/81%): on sparse point clouds, extra views add noise faster than evidence unless aggregation is vote-based. The cost of voting is a larger *plausible* bucket — split votes often land on *clutter* — trading decisiveness for safety, the right trade for an autonomously written robot database.

TABLE II
MANUAL AUDIT OF FINAL TYPE LABELS (57 OBJECTS). C/P/W = CORRECT / PLAUSIBLE / WRONG; STRICT = C, LENIENT = C+P.

Configuration	C	P	W	strict	lenient
Uni3D only (no VLM)	28	5	24	49%	58%
1 view, base prompt, sparse	34	12	11	60%	81%
4 views early fus., sparse	28	10	19	49%	67%
1 view, guardrails, dense	29	11	17	51%	70%
4 views early fus., dense	26	12	19	46%	67%
4 views late fus., dense	28	14	15	49%	74%

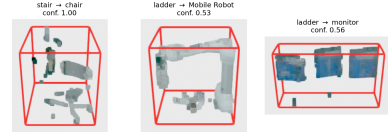


Fig. 4. Representative late-fusion corrections (one of four dense views; provisional label $\rightarrow$ fused label, vote-share confidence). The 3D classifier’s industrial bias (*stair*, *ladder*) is corrected to furniture, robots, and displays; the chair correction is detailed view-by-view in Fig. 2.

(3) *Vote unanimity is a free precision signal*. 17 of 57 objects received identical labels from all four views; 16 of those (94%) were audited correct, versus 30% among split-vote objects. Gating on the stored `viewVotes` thus gives a downstream consumer a calibrated route-to-human-review policy that per-call confidence alone did not provide.

Render density cut both ways, and produced the study’s starkest warning: dense renders made four dotted-grid panels look so convincingly like keyboards (per-view confidence up to 0.97) that the label passed our render-based audit — only the facility owner’s inventory check revealed it was wrong, and Table II grades every such label wrong. The incident is the strongest argument for the gating above: a label that fools both the VLM and a human at the same renders was still caught by vote splits (these objects never reached unanimity). Run-to-run sampling variance (± 2 –3 objects) bounds how finely the configurations can be ranked.

C. Implicit Model and Completed Records

On the late-fusion output, Prompt 2 marked 6 of 57 objects as key objects (control panels, a large machine, mobile robots, a wall display) and 50 as movable. Fig. 6 shows two completed TOSM records as stored in the output message, including the audit trail (per-view votes, fused confidence, one-sentence reason). Mean fused confidence was 0.71, and the 17 unanimous objects carry confidence 1.0 by construction, aligning the stored confidence with the observed 94% unanimity precision. By design the generated records are a reviewable draft, not ground truth: the semantic model remains user-editable and every amendment is logged in the record’s audit trail. An owner review exercised exactly this loop on the implicit model, correcting two movability entries (a wheeled machine judged fixed, a wall-mounted display judged movable) at negligible cost.

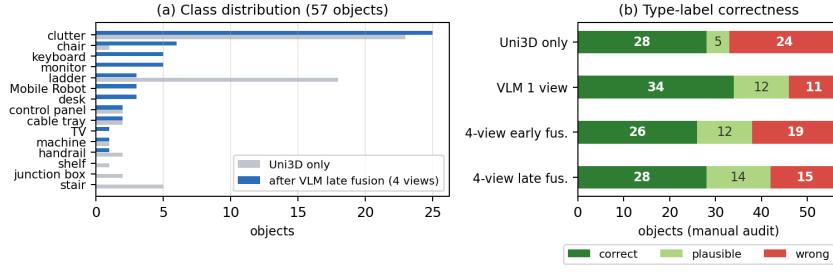


Fig. 5. Effect of VLM verification on the 57-object scene. (a) Class distribution, classifier-only vs. after late fusion: implausible *ladder/stair* populations are redistributed to displays and furniture (the keyboard outputs were later rejected by the on-site inventory, Sec. IV-B). (b) Manual audit: early fusion degrades single-view accuracy; late fusion is the strongest multi-view strategy (15 wrong labels vs. 19 for early fusion).

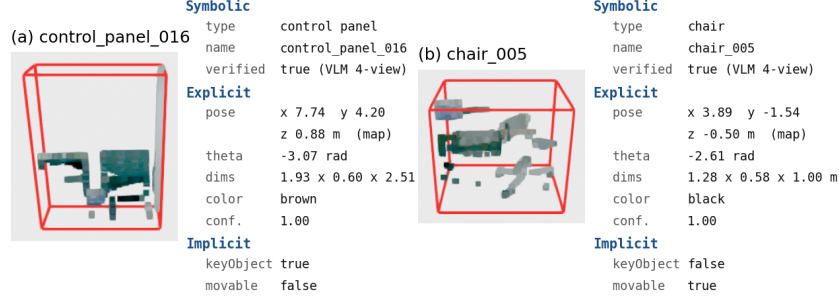


Fig. 6. Completed TOSM semantic-object records (dense render + symbolic / explicit / implicit fields) as serialized into the VDA 5050-style message. (a) A control panel, unanimously verified and marked as a fixed key landmark. (b) An office chair corrected from *stair*, movable and not a landmark.

D. Discussion and Limitations

The division of labor is the main design point: everything measurable stays deterministic and VLM-immutable, so hallucination can corrupt neither poses nor dimensions, and the VLM contributes exactly the judgments geometry cannot make — naming and affordances — in an auditable form (per-view votes, forced-tool JSON, stored reasons). All 741 tool calls across the five VLM configurations returned schema-valid JSON. Remaining errors concentrate where the render itself is ambiguous (texture-poor slabs, merged clusters, sparse strips), pointing to upstream fixes — finer segmentation, mesh renders — not to the fusion protocol. Absolute strict accuracy (~49–60%) reflects a deliberately hard test bed: a cluttered showroom scanned once, without manual cleanup.

V. CONCLUSION

We presented an automated pipeline that turns a single terrestrial laser scan into TOSM semantic-object records delivered as a VDA 5050-style JSON message, using a VLM in a tightly scoped, schema-forced role. A six-configuration study showed that the intuitive multi-view prompt — all views at once — degrades verification on sparse point renders, whereas per-view classification with confidence-weighted late fusion is the strongest multi-view strategy: wrong labels drop from 24 (classifier-only) to 15, lenient accuracy reaches 74%, and vote unanimity provides a 94%-precise gating signal at \$2.6 per scene. Future work includes mesh-based renders, finer segmentation, and coupling the records to place modeling and a live digital twin.

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